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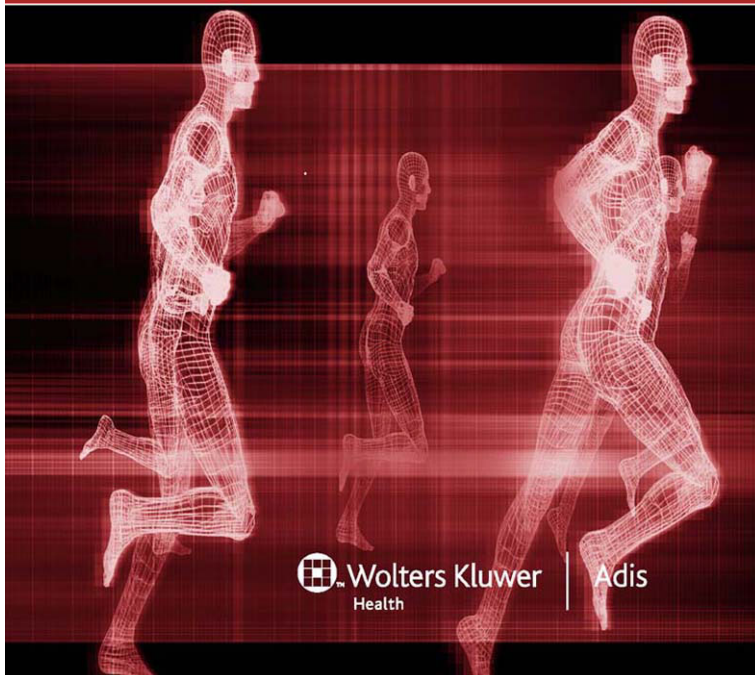


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Lactate Threshold Concepts

How Valid are They?

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Abstract

During the last nearly 50 years, the blood lactate curve and lactate thresholds (LTs) have become important in the diagnosis of endurance performance. An intense and ongoing debate emerged, which was mainly based on terminology and/or the physiological background of LT concepts. The present review aims at evaluating LTs with regard to their validity in assessing endurance capacity. Additionally, LT concepts shall be integrated within the ‘aerobic-anaerobic transition’ – a framework which has often been used for performance diagnosis and intensity prescriptions in endurance sports.

Usually, graded incremental exercise tests, eliciting an exponential rise in blood lactate concentrations (bLa), are used to arrive at lactate curves. A shift of such lactate curves indicates changes in endurance capacity. This very global approach, however, is hindered by several factors that may influence overall lactate levels. In addition, the exclusive use of the entire curve leads to some uncertainty as to the magnitude of endurance gains, which cannot be precisely estimated. This deficiency might be eliminated by the use of LTs.

The aerobic-anaerobic transition may serve as a basis for individually assessing endurance performance as well as for prescribing intensities in

endurance training. Additionally, several LT approaches may be integrated in this framework. This model consists of two typical breakpoints that are passed during incremental exercise: the intensity at which bLa begin to rise above baseline levels and the highest intensity at which lactate production and elimination are in equilibrium (maximal lactate steady state [MLSS]).

Within this review, LTs are considered valid performance indicators when there are strong linear correlations with (simulated) endurance performance. In addition, a close relationship between LT and MLSS indicates validity regarding the prescription of training intensities.

A total of 25 different LT concepts were located. All concepts were divided into three categories. Several authors use fixed bLa during incremental exercise to assess endurance performance (category 1). Other LT concepts aim at detecting the first rise in bLa above baseline levels (category 2). The third category consists of threshold concepts that aim at detecting either the MLSS or a rapid/distinct change in the inclination of the blood lactate curve (category 3).

Thirty-two studies evaluated the relationship of LTs with performance in (partly simulated) endurance events. The overwhelming majority of those studies reported strong linear correlations, particularly for running events, suggesting a high percentage of common variance between LT and endurance performance. In addition, there is evidence that some LTs can estimate the MLSS. However, from a practical and statistical point of view it would be of interest to know the variability of individual differences between the respective threshold and the MLSS, which is rarely reported.

Although there has been frequent and controversial debate on the LT phenomenon during the last three decades, many scientific studies have dealt with LT concepts, their value in assessing endurance performance or in prescribing exercise intensities in endurance training. The presented framework may help to clarify some aspects of the controversy and may give a rationale for performance diagnosis and training prescription in future research as well as in sports practice.

1. Historical Remarks on Endurance Performance Diagnosis

As early as 1808, Berzelius observed that lactic acid was produced in the muscles of hunted stags.^[1] About a century later, several scientists studied the biochemistry of energy metabolism and muscle contraction in more detail. This led to a much deeper understanding of the formation of lactic acid (lactate and hydrogen ions) during intense exercise.^[2-5] At that time, it was common belief that lactic acid is a waste product of glycolysis and will be formed when oxygen delivery to exercising muscles is not sufficient and muscle anaerobiosis occurs.^[2,6,7] This view has been challenged considerably during the last two dec-

ades. Anaerobic glycolysis and, thus, lactate kinetics rather seem to be an ongoing process – even in the resting individual – which is highly related to the metabolic rate but not necessarily to oxygen availability (for detailed review see Gladden,^[1,8] Brooks,^[9] Robergs et al.^[10]).

In the first half of the 20th century the concept of maximum oxygen consumption as the first and probably most common means of evaluating aerobic endurance capacity was developed by the working group of Nobel Laureate AV Hill.^[6] maximal oxygen uptake ($\dot{V}O_{2\max}$) has been established as a valuable tool to distinguish between fit and unfit subjects. However, several concerns were raised regarding the sensitivity of $\dot{V}O_{2\max}$. For instance, it is difficult to discriminate

between subjects of homogenous performance levels by means of $\dot{V}O_{2\max}$.^[11-18] In addition, sufficient effort during whole-body work and, therefore, adequate motivation of the investigated subject is necessary to appropriately determine $\dot{V}O_{2\max}$. Particularly in clinical settings with diseased patients, whole-body exhaustion is difficult to attain or is even avoided because of the risk of adverse events.^[19,20]

Therefore, attempts have been made to establish sub-maximal parameters to assess cardio-respiratory fitness in patients and athletes. Early research by the working group of Hollmann established the so-called 'point of optimum ventilatory efficiency' corresponding to the first increase in the ventilatory equivalent of oxygen and of arterial lactate concentrations during incremental exercise.^[19,21] A few years later, Wasserman and McIlroy^[22] determined this intensity by plotting ventilation versus oxygen uptake in cardiac patients and named it the 'anaerobic threshold' (LT_{An}). At that time, routine determination of blood lactate concentrations (bLa) was associated with several difficulties and gas exchange measurements were more common – especially in clinical settings. Therefore, it became popular to detect the LT_{An} by means of gas exchange analysis.

In the 1960s, the enzymatic method for measuring lactate concentrations from capillary blood samples was developed. This led to the increasing popularity of using bLa as a parameter to assess endurance capacity as well as for classifying work rate during exercise.^[19,23,24] In the following years, numerous lactate threshold (LT) concepts were developed. The number of scientific studies on LTs has increased enormously up to now and the sub-maximal course of bLa during incremental exercise has probably become one of the most important means in the diagnosis of endurance performance in sports practice.^[15,16,25,26] However, the variety of different threshold concepts has led to considerable confusion and misinterpretation.

An intense and ongoing debate emerged, which was mainly based upon terminology and/or the physiological background of LT concepts.^[27] Early assumptions on lactate produc-

tion and distribution in the organism have been challenged.^[1,8-10,28] It has been argued that bLa increase continuously rather than show a clear threshold during incremental exercise. Furthermore, the contribution of aerobic and anaerobic pathways to energy production does not change suddenly but shows a continuous transition and, therefore, the term 'threshold' might be misleading.^[29]

Against this background and to unravel the confusion, it seems valuable to give a summary on published LT concepts. The present review is mainly aimed at evaluating the located LT concepts with regard to their validity in assessing aerobic endurance capacity and prescribing training intensity. A further aim was to try to integrate those concepts into a framework that was originally called the aerobic-anaerobic transition.^[30-32]

It has to be emphasized that this text focuses on LTs only. Although a close link between lactate and gas exchange markers has often been proposed,^[21,31,33-36] there is still controversial debate with regard to the underlying physiological mechanisms.^[37] A comprehensive review on gas exchange thresholds has recently been published.^[31] Additionally, it is not within the scope of this article to exhaustively review the biochemistry of glycolysis and lactate metabolism.

2. Incremental Exercise Testing and the Interpretation of Blood Lactate Curves

2.1 The Entire Blood Lactate Curve

Usually, graded incremental exercise tests (GXTs) are used to evaluate aerobic endurance performance capacity. Typically, an exponential rise in bLa during incremental exercise testing can be observed (figure 1). The issue of interest is to interpret the resulting lactate curve with regard to endurance capacity. It is generally accepted that a rightward shift of the lactate curve (lower bLa at given workload) can be interpreted in terms of an improved endurance capacity^[38-40] and, in contrast, a shift to the left (higher bLa at given workload) is usually considered to represent worsening endurance capacity.^[41]

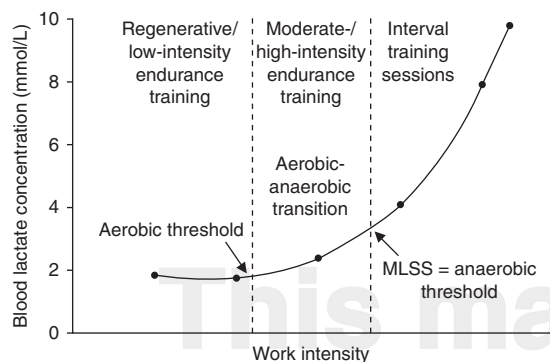


Fig. 1. A typical lactate-workload plot including the aerobic-anaerobic transition as a framework to derive endurance training intensities for different intensity zones. **MLSS** = maximal lactate steady state.

Overall lactate levels are known to be influenced by depleted glycogen stores (due to a low carbohydrate diet or preceding exhaustive exercise).^[42-44] For instance, lower bLa at the same work rates have been reported in a glycogen-depleted subject compared with a subject in normal condition. This may lead to a downward shift of the lactate curve and it is important that this is not falsely interpreted as an enhancement in endurance capacity.^[45] Furthermore, several other factors like muscle fibre composition, glycolytic and lipolytic enzyme activity as well as capillary or mitochondrial density might influence blood lactate curves.^[46] Additionally, the entire lactate curve is dependent on several other methodological issues, which should be taken into account when interpreting test results.

2.1.1 Test Design and Data Treatment

It is of note that the specific GXT protocol can vary considerably with regard to starting and subsequent work rates, work rate increments and stage duration. A recent review focused on the influence of varying test protocols on markers usually used in the diagnosis of endurance performance.^[47] For instance, varying stage duration or work rate increments may lead to relevant differences in blood lactate curves and LTs.^[48-50] A possible reason might be the time allowed for

lactate diffusion in the blood until the next work rate increment.^[47]

In addition, there has been great debate on the best fitting procedure for the obtained bLa data set. For instance, a single-^[51] or double-phase model^[52] using two or three linear regression segments, a double-log model,^[53] a third-order polynomial^[54] or an exponential function^[55] have been used in previous studies. Up to now, no generally accepted fitting procedure has been established.^[47] Thus, it seems appropriate that test design as well as data fitting procedures should be chosen (and reported) as has been originally described for a certain LT.

2.1.2 Methodology of Blood Lactate Determination

From a methodological point of view, the site (earlobe, fingertip) as well as the method (venous, arterial, capillary) of blood sampling^[56,57] and the laboratory methods (lactate analyser, analysed blood medium)^[58-60] may also affect the test result. Samples taken from the earlobe have uniformly been shown to result in lower bLa than samples taken from the fingertip.^[57,61,62] With regard to the analysed blood medium, plasma values were considerably higher than whole venous lactate concentrations, with capillary values lying in between.^[48,56,63-65] In addition, several studies reported partly considerable differences between various lactate analysers (portable field vs laboratory analysers, amperometric vs photometric method) and under various climatic conditions.^[58,66-69]

The analysis of the whole blood lactate curve is a very global approach to evaluating endurance capacity. On the one hand, this approach is affected by the above-mentioned factors on overall lactate levels. On the other hand, the use of the entire curve leads to some uncertainty as to the magnitude of endurance gains that cannot be precisely estimated. However, the use of LTs enables a quantitative evaluation of changes in endurance performance. In addition, the ideal LT concept would not be affected by the above-mentioned factors. There is evidence that approaches that analyse relative changes in bLa during GXTs may be favourable compared with the use of absolute lactate values in this regard.^[56,67]

2.2 A Framework for Endurance Diagnosis and Training Prescriptions

In 1979, Kindermann et al.^[30] introduced the concept of the aerobic-anaerobic transition as a framework for performance diagnosis and training prescription in endurance sports (figure 1). Since then, this framework has been adopted, applied and refined by several scientists either using lactate or gas exchange markers.^[16,26,31,33,34,46,70-75]

This model consists of two typical breakpoints that are passed during incremental exercise. In the low intensity range, there is an intensity at which bLa begin to rise above baseline levels. This intensity was originally determined using gas exchange measurements,^[21,22] and Wasserman called it the 'anaerobic threshold'. This term has since been used for various LTs, particularly those with a different physiological background,^[33,75] and, thus, has caused considerable confusion. Kindermann et al.^[30] and Skinner and McLellan^[34] suggested this intensity be called the 'aerobic threshold' (LT_{Aer}), because it marks the upper limit of a nearly exclusive aerobic metabolism and allows exercise lasting for hours. This intensity might be suitable for enhancing cardio-respiratory fitness in recreational sports, for cardiac rehabilitation in patients or for low-intensity and regenerative training sessions in high level endurance athletes.^[16,25,26,32,70,76-81]

Exercise intensities only slightly above the LT_{Aer} result in elevated but constant bLa during steady-state exercise and can be maintained for prolonged periods of time (~4 hours at intensities in the range of the first increase in bLa^[82-84] and 45–60 minutes at an intensity corresponding to the maximal lactate steady state [MLSS]^[85,86]). Although anaerobic glycolysis is enhanced, it is speculated that such intensities may induce a considerable increase in the oxidative metabolism of muscle cells.^[30,87] Theoretically, a high stimulation of oxidative metabolism for as long a period of time as is possible in this intensity range might be an appropriate load for endurance training. The highest constant workload that still leads to an equilibrium between lactate production and lactate elimination represents the MLSS.

Some authors suggested that this intensity be called the 'anaerobic threshold'.^[27,30,49,88]

It has been shown that the constant bLa at MLSS is not equal in all individuals and can vary considerably (values from 2 up to 10 mmol/L were reported in several studies).^[50,72,86,89-93] Beneke and von Duvillard^[94] as well as Beneke et al.^[95] reported that bLa at MLSS is dependent on the motor pattern of exercise. Therefore, it was suggested that to determine the LT_{An} , individualized approaches rather than a fixed bLa should be used.^[88,96,97]

The MLSS represents the upper border of constant load endurance training.^[30,49,71,95] Intensities above the MLSS have been used to guide interval training sessions in different endurance sports.^[26,31,98-102]

The intensity range between LT_{Aer} and LT_{An} is called the aerobic-anaerobic transition. The described thresholds (first increase in bLa and MLSS) have recently also been called 'lactate threshold and lactate turnpoint', 'lactate threshold and anaerobic threshold', or 'anaerobic threshold 1 and 2', respectively.^[26,75,103,104] Within the present review, it was decided to stick to the originally introduced nomenclature.^[30,31,34]

There has been an exhaustive debate whether there exist clear breakpoints in the lactate/work rate relationship or whether lactate increase is rather a continuous function during incremental work.^[47] Furthermore, the terms 'aerobic' and 'anaerobic' threshold may suggest clearly discernible physiological processes. However, these processes are rather of a transitional nature with aerobic and anaerobic energetic pathways always simultaneously contributing to energy production during both low- and high-intensity exercise. However, the proposed model seems appropriate both from a practical and from a didactical point of view. In addition, there is evidence that the described breakpoints may have some exercise physiological relevance. It has been shown that exercise above the MLSS is associated with an over-proportional excretion of stress hormones as well as of immunological markers during constant load exercise.^[105,106] Furthermore, Lucia et al.^[107] observed changes in electromyographical activity of the *vastus lateralis* and

rectus femoris that were coincidental with the aerobic-anaerobic transition in 28 elite male cyclists.

The widespread use of this model as well as the absence of an accepted alternative was the rationale for using this framework in the present review to categorize published LT concepts.

3. Validation of Lactate Thresholds

3.1 Competition Performance

It is widely accepted that LTs (and the submaximal course of bLa during incremental exercise) are a criterion measure for aerobic endurance performance.^[24,26,30,72,81,108] In particular, it has been shown that LTs are superior to maximal oxygen uptake when assessing endurance performance in homogenous groups of athletes.^[11,12,109-111] The obvious gold standard to validate an LT concept is to compare it with the most recent competition performance in an endurance event (concurrent validity) or to assess its value in predicting endurance performance in future events (predictive validity). As an alternative to competition performance, the results of laboratory tests simulating an endurance event can be used. This might have the advantage of a higher standardization and, therefore, these test results may be more reliable. Correlations between the test value (LT) and the validity criterion (competition performance) can be dependent on several confounding factors such as, for example, the chosen competitive event (duration, laboratory or outdoor, athletic track or off-road), the sport that is evaluated as well as sex or age group and its heterogeneity in terms of endurance.

3.2 The Maximal Lactate Steady State

Endurance capacity can – from a metabolic point of view – be regarded as the highest steady state by energy supply from oxidative phosphorylation.^[87] Therefore, another approach to assess aerobic endurance performance is the determination of the highest constant exercise intensity that can be maintained for a longer period of time

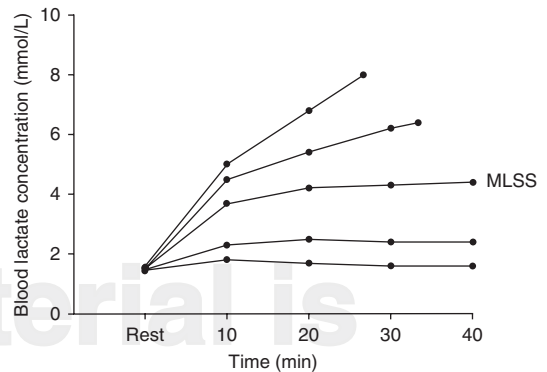


Fig. 2. The blood lactate response to several constant workload exercises with different intensities. The highest workload during which blood lactate concentrations can be still accepted as being steady state is defined as the maximal lactate steady state (MLSS).

without a continuous rise in bLa. This intensity represents the MLSS, which has been shown to be highly related to competition performance in endurance events (r [correlation coefficient] = 0.92 with 8 km running, $r = 0.87$ with 5 km running and $r = 0.84$ with 40 km cycling time trial speed, respectively).^[112-114] The MLSS has been defined by some authors as the ‘anaerobic threshold’ because it represents an exercise intensity that can be maintained without considerable contribution of anaerobic metabolism.^[27,30,50,72,115] Each higher intensity results in a clearly identifiable increase in bLa with time during constant load work.^[50,86,88]

The gold standard for the determination of the MLSS is performing several constant load trials of at least 30 minutes’ duration on different days at various exercise intensities (in the range of 50–90% $\dot{V}O_{2max}$, figure 2).^[49,50,86,116,117] An increase in bLa of not more than 1 mmol/L between 10 and 30 minutes during the constant load trials appears to be the most reasonable procedure for MLSS determination.^[86,115]

MLSS represents a steady state in several but not all physiological parameters. For instance, oxygen uptake, carbon dioxide output, respiratory exchange ratio and bicarbonate concentration were reported to remain nearly constant during constant load exercise at MLSS, but respiratory rate and heart rate significantly increased during this time.^[85,118]

In several endurance sports it is recommended to aim at a defined metabolic strain when a certain training stimulus is intended.^[71,73,119,120] Therefore, it seems likely that training intensities for endurance training can be appropriately described when MLSS is known.

For the purposes of this review based on the above-mentioned rationales, LTs are considered valid as performance indicators when there are high linear correlations with (simulated) endurance performance. In addition, a close relationship between LTs and MLSS suggests validity with regard to the prescription of training intensities. Therefore, it is desirable that LTs should fulfil both validity criteria.

4. Lactate Threshold Concepts

For the purposes of the present paper, the MEDLINE database PubMed was searched for the search terms ‘lactate threshold’, ‘aerobic threshold’ and ‘anaerobic threshold’ combined with either ‘endurance performance’ or ‘maximal lactate steady state’. Additionally, the references of the selected articles were searched for further relevant papers. The located original publications were searched for papers describing different LT concepts (section 4.1), a correlation between LTs and (simulated) endurance performance (section 4.2) or the relationship between LTs and the MLSS (section 4.3).

4.1 Located Lactate Threshold Concepts

A total of 25 different LT concepts were located. Two studies were excluded from the present analysis because threshold determination was not solely based on bLa but also took gas exchange measurements into account.^[121,122] All threshold concepts were divided into three categories. Several authors used so-called fixed blood lactate thresholds (LT_{fix}) during incremental exercise to evaluate aerobic endurance performance. These fixed bLas were set at 2, 2.5, 3 or 4 mmol/L^[24,108,123-125] with LT4 (4 mmol/L lactate threshold, originally described by Mader et al.^[24] and by others later as the onset of blood

lactate accumulation [OBLA]^[108]) being the most frequently used method.

4.1.1 Aerobic Lactate Thresholds

Table I shows an overview of LT concepts that could be categorized as the first rise in bLa above baseline levels (LT_{Aer}). Several researchers described the procedure to determine this threshold with terms like “the first significant/marked/systematic/non-linear/sharp/abrupt sustained increase in bLa above baseline”.^[30,110,126-133,138] Although the visual determination of the first rise of bLa above baseline levels seems obvious and simple, in practice it is associated with considerable problems because of the only slight changes in bLa on the first steps during GXTs. Yeh et al.^[142] demonstrated that the visual detection of the LT_{Aer} (in that study called ‘anaerobic threshold’) led to relevant differences between observers. Therefore, it does not seem appropriate to determine this threshold by simple visual inspection. Thus, other methods were developed to make threshold determination more objective. For instance, some authors took the typical error of their lactate analysers into account and

Table I. Lactate threshold concepts that were categorized in the aerobic threshold category. For further explanation see text

Method and description
Work intensity or oxygen uptake
before/at which bLa begins to increase above baseline level ^[110,126]
at which bLa exhibits a marked/systematic/significant/non-linear/sharp/abrupt sustained increase above baseline value ^[30,110,127-133]
first significant elevation of lactate level (approximately 2 mmol/L) ^[30,34]
before an elevation in bLa above baseline (at least 0.2 mmol/L due to error of lactate analyser) ^[123,134]
rise in delta lactate (onset of plasma lactate accumulation) ^[109]
at minimum lactate equivalent (bLa divided by oxygen uptake or work intensity) ^[36,135-137]
at which plasma lactate concentration begins to increase when log bLa is plotted against log (work intensity) ^[53]
at which bLa increases 0.5 mmol/L above resting concentration ^[138]
at which bLa increases 1 mmol/L above baseline (i.e. lactate at low intensity corresponding to 40–60% VO _{2max}) ^[111,139]
preceding a bLa increase by 1 mmol/L or more ^[140,141]
bLa =blood lactate concentrations; VO_{2max} =maximal oxygen uptake.

Table II. Lactate threshold concepts that were categorized in the anaerobic threshold category. For further explanation see text

Threshold concept	Method and description
IAT (Stegmann et al.) ^[88]	Tangent to bLa curve from recovery curve where bLa is equal to the value at end of GXT
IAT (Keul et al.) ^[96]	Tangent to bLa curve at 51°
IAT (Simon et al.) ^[97]	Tangent to bLa curve at 45°
IAT (Berg et al.) ^[137]	Intersection point between tangent for the minimum lactate equivalent and the linear function for the final 90 sec of GXT
IAT (Bunc et al.) ^[143]	Intersection between the exponential regression of the lactate curve and the bisector of the tangents of the upper and lower parts of the lactate curve
IAT (Dickhuth et al.) ^[36,136]	1.5 mmol/L above minimum lactate equivalent
IAT (Baldari and Guidetti) ^[144]	The second lactate increase of at least 0.5 mmol/L from the previous value
D _{max} (Cheng et al.) ^[54]	Maximal distance from bLa curve to the line formed by its endpoints
D _{mod} (Bishop et al.) ^[140]	Maximal distance from bLa curve to the line formed by the point before the first rise in bLa and the value at cessation of exercise
Lactate turnpoint ^[103]	The final running velocity before the observation of a sudden and sustained increase in bLa between LT _{Aer} and $\dot{V}O_{2max}$
Lactate minimum speed ^[145]	Minimum in bLa during GXT after high intensity exercise

bLa = blood lactate concentration; **GXT** = incremental exercise test; **IAT** = individual anaerobic threshold; **LT_{Aer}** = aerobic threshold; **$\dot{V}O_{2max}$** = maximal oxygen uptake.

determined this LT as the workload 0.2 mmol/L above the lowest exercise lactate value.^[123] Hughson and Green^[138] arbitrarily chose an increase of 0.5 mmol/L above resting lactate concentrations. Another work group^[111,139] chose a 1 mmol/L increment above lactate levels at low intensity (~40% to 60% $\dot{V}O_{2max}$) because it could be determined objectively and in a standardized manner in all subjects. Furthermore, the lowest value when bLa is divided by work intensity or $\dot{V}O_2$ has also been used as a marker for LT_{Aer} (minimum lactate equivalent).^[36,135-137] Whereas Beaver and colleagues^[53] used a log-log transformation to assess the first rise in bLa more objectively as the intersection of two linear regressions, Farrell et al.^[109] plotted the difference in bLa between two consecutive stages against work intensity and determined the first rise of this relationship (onset of plasma lactate accumulation).

4.1.2 Anaerobic Lactate Thresholds

All threshold concepts that were assigned either to the MLSS or to a rapid/distinct change in the inclination of the blood lactate curve were categorized as LT_{An} (table II).

Originally, the LT4 was established because it seemed to be the highest bLa that was sustainable for a longer duration and, therefore, was regarded

as the upper border for constant load endurance training.^[24] It was soon recognized that a fixed bLa does not take into account considerable interindividual differences and that LT4 may frequently underestimate (particularly in anaerobically trained subjects) or overestimate (in aerobically trained athletes) real endurance capacity.^[88,96,97,146] Therefore, several so-called 'individualized' LT concepts were developed. For instance, Keul et al.^[96] and Simon et al.^[97] determined the individual anaerobic threshold (IAT) at a certain inclination of the lactate curve (tangent of 51° and 45°, respectively). However, it seems questionable whether the use of a fixed inclination may reflect individual lactate kinetics better than a fixed bLa.

Stegmann et al.^[88] developed a more complicated model that is based on the exercise lactate curve as well as on the lactate course during the early recovery period. This model is based on several assumptions regarding lactate distribution in blood and muscle compartments, lactate diffusion through biological membranes and lactate elimination. However, some of these premises have been challenged.^[8,147]

Berg et al.^[137] defined the LT_{An} as the intersection point between the tangent at the minimum lactate equivalent and the linear function

for the final 90 seconds of GXT. Similarly, Bunc et al.^[143] determined the LT_{An} as the intersection between the exponential regression of the lactate curve and the bisector of the tangents on the upper and lower parts of the regression. A comparable model was established by Cheng et al.^[54] and called the D_{max} method. Those authors determined the maximal perpendicular distance of the lactate curve from the line connecting the start with the endpoint of the lactate curve. It is obvious that these threshold models are dependent on the start intensity as well as the maximal effort spent by the subjects. To eliminate the influence of the start point of the GXT, Bishop et al.^[140] connected the LT_{Aer} with the endpoint of the lactate curve and observed that this modified D_{max} threshold (D_{mod}) was also highly correlated with performance during a 1-hour time trial in 24 female cyclists.

Tegtbur et al.^[145] developed the so-called lactate minimum test. This test starts with a short supramaximal exercise leading to high bLa. A short rest period (about 8 minutes)^[145] should allow for an equilibrium between muscle and bLa. Immediately after this rest period, a standard incremental exercise test is conducted. After an initial fall of bLa at low exercise intensities, bLa begins to rise again. The lowest point of the lactate curve, the lactate minimum speed (LMS), is assumed to mark the LT_{An} . This procedure has recently been criticized because standardization is difficult.^[112,148] For instance, the induced acidosis prior to the incremental test is unlikely to be uniform for different subjects. Additionally, initial intensity as well as stage increment and duration seem to affect LMS. Furthermore, supramaximal exercise might be contraindicated in some instances, for example in cardiac patients or in athletes at some time points during their training.

Baldari and Guidetti^[144] defined the IAT as the workload corresponding to the second lactate increase of at least 0.5 mmol/L with the second increase greater than or equal to the first one. A limitation to this approach is that only discrete stages according to the test protocol can be identified as threshold workload. Additionally, those authors determined the IAT by plotting each lactate value against the preceding work-

load. This was claimed to be done because during 3-minute stages no steady-state lactate level could be reached^[147] and, therefore, it was hypothesized that a lactate value at a given 3-minute stage would represent the realistic value of the previous stage.

From empirical observations, the work group of Dickhuth et al.^[36,135,136] estimated the IAT at a blood lactate concentration 1.5 mmol/L above the minimum lactate equivalent (i.e. above LT_{Aer}). Finally, the lactate turnpoint (LTP) has been defined as the final running velocity before the observation of a sudden and sustained increase in bLa between LT_{Aer} and $\dot{V}O_{2max}$.^[103]

Reproducibility of the velocity or power output at LTs has been reported to be high ($r > 0.9$, independent of whether LT_{fix} , LT_{Aer} or LT_{An} were analysed).^[52,149-152] For $\dot{V}O_2$ at LTs, reliability coefficients seem to be slightly lower ($r = 0.8-0.9$).^[150,152,153]

4.2 Lactate Thresholds and (Simulated) Competition Results

Thirty-eight studies were located that compared LT values with performance in endurance events or simulated competitions. Six studies were excluded from the analysis. Three of these studies compared an LT obtained during cycling exercise with running performance,^[110,154,155] two studies only reported LT as a fraction of $\dot{V}O_{2max}$,^[11,156] and one study reported correlations with time-to-exhaustion in an open-end interval programme.^[157] A total of 32 studies were thus included in this analysis.

Eighteen studies evaluated the correlation of the work intensity (running velocity or $\dot{V}O_2$) at various LTs with performance in running competitions of different distances (800 m up to marathon; table III).^[108,109,112,123,124,129-132,134,135,158-164] Competition distances from 0.8 to 3.2 km, from 5 km to 16.1 km and from 21.1 to 42.2 km were subsumed as correlates of short-, middle- and long-distance endurance events. The main result was that nearly all studies reported high correlation coefficients with (simulated) competition results. These results were confirmed by Weltman

Table III. Correlation coefficients between lactate thresholds and running performance over various distances

Threshold concept	0.8–3.2 km		5 km–16.1 km		19.3–42.2 km	
	v	$\dot{V}O_2$	v	$\dot{V}O_2$	v	$\dot{V}O_2$
LT _{fix}	0.82 ^[135]	0.79 ^[123]	0.88 ^[135]	0.90 ^[159]	0.91 ^[135]	0.76 ^[161]
	0.88 ^[123]	0.75 ^[123]	0.91 ^[135]	0.92 ^[159]	0.81 ^[135]	0.83 ^[163]
	0.86 ^[123]	0.75 ^[123]	0.91 ^[159]	0.92 ^[159]	0.98 ^[124]	0.73 ^[163]
	0.85 ^[123]	0.72 ^[134]	0.93 ^[159]	0.83 ^[159]	0.98 ^[124]	
	0.87 ^[134]	0.74 ^[134]	0.91 ^[159]	0.88 ^[159]	0.98 ^[124]	
	0.85 ^[134]	0.75 ^[134]	0.84 ^[159]	0.93 ^[159]	0.68 ^[129]	
	0.84 ^[134]	0.73 ^[158]	0.91 ^[159]	0.86 ^[163]	0.96 ^[108]	
	0.93 ^[158]	0.60 ^[132]	0.94 ^[159]	0.74 ^[163]	0.91 ^[163]	
	0.78 ^[132]	0.51 ^[131]	0.83 ^[160]		0.92 ^[163]	
	0.68 ^[131]	0.55 ^[131]	0.81 ^[112]			
	0.85 ^[131]	0.69 ^[131]	0.95 ^[163]			
	0.88 ^[131]		0.94 ^[163]			
Median (min–max)	0.85 (0.68–0.93)	0.73 (0.51–0.79)	0.91 (0.81–0.95)	0.89 (0.74–0.93)	0.92 (0.68–0.98)	0.76 (0.73–0.83)
LT _{Aer}	0.74 ^[135]	0.77 ^[123]	0.73 ^[135]	0.89 ^[109]	0.76 ^[135]	0.91 ^[109]
	0.85 ^[123]	0.61 ^[134]	0.79 ^[135]	0.91 ^[109]	0.81 ^[135]	0.89 ^[109]
	0.70 ^[134]	0.84 ^[158]	0.78 ^[160]	0.84 ^[162]	0.78 ^[129]	0.69 ^[163]
	0.93 ^[158]	0.69 ^[132]	0.96 ^[109]	0.83 ^[162]	0.97 ^[109]	0.52 ^[163]
	0.77 ^[132]	0.77 ^[131]	0.97 ^[109]	0.79 ^[162]	0.98 ^[109]	0.66 ^[163]
	0.43 ^[131]	0.66 ^[131]	0.79 ^[130]	0.69 ^[162]	0.90 ^[163]	0.42 ^[163]
	0.65 ^[131]	0.64 ^[131]	0.83 ^[130]	0.92 ^[162]	0.91 ^[163]	0.80 ^[163]
	0.70 ^[131]	0.85 ^[109]	0.79 ^[130]	0.79 ^[162]	0.87 ^[163]	0.65 ^[163]
	0.91 ^[109]	0.62 ^[162]	0.84 ^[130]	0.76 ^[130]	0.86 ^[163]	
		0.66 ^[162]	0.83 ^[130]	0.77 ^[130]	0.83 ^[163]	
		0.58 ^[162]	0.81 ^[130]	0.84 ^[130]	0.77 ^[163]	
			0.93 ^[112]	0.81 ^[130]		
			0.94 ^[163]	0.82 ^[130]		
			0.92 ^[163]	0.88 ^[130]		
			0.92 ^[163]	0.72 ^[163]		
			0.89 ^[163]	0.56 ^[163]		
			0.87 ^[163]	0.66 ^[163]		
			0.85 ^[163]	0.52 ^[163]		
				0.81 ^[163]		
				0.69 ^[163]		
Median (min–max)	0.74 (0.43–0.93)	0.66 (0.58–0.85)	0.84 (0.73–0.97)	0.79 (0.45–0.92)	0.86 (0.76–0.98)	0.68 (0.42–0.91)
LT _{An}	0.88 ^[135]		0.91 ^[135]	0.83 ^[163]	0.93 ^[135]	0.68 ^[161]
			0.92 ^[135]	0.70 ^[163]	0.93 ^[135]	0.83 ^[163]
			0.86 ^[160]	0.81 ^[163]	0.90 ^[163]	0.71 ^[163]
			0.83 ^[112]	0.66 ^[163]	0.91 ^[163]	0.81 ^[163]
			0.93 ^[163]	0.45 ^[164]	0.90 ^[163]	0.67 ^[163]
			0.91 ^[163]	0.45 ^[164]	0.89 ^[163]	
			0.94 ^[163]			
			0.90 ^[163]			
			0.76 ^[164]			
			0.73 ^[164]			
Median (min–max)	0.88		0.91 (0.83–0.94)	0.76 (0.66–0.83)	0.91 (0.89–0.93)	0.71 (0.67–0.83)

LT_{fix}=fixed lactate threshold; LT_{Aer}=aerobic threshold; LT_{An}=anaerobic threshold; v=velocity; $\dot{V}O_2$ =oxygen uptake.

et al.,^[123,134] who cross-validated the obtained regression equations and found high correlation coefficients between actual and predicted scores. There is a tendency for higher correlations with longer endurance events (0.43–0.93 in short-term

events vs 0.68–0.98 over the long-distance competitions). Additionally, correlations tended to be higher for LT_{fix} and LT_{An} compared with LT_{Aer}. This might be due to the average intensity in running events being higher than the intensity

corresponding to the first increase in bLa. In total, the results of the analysed studies point to a common variance of LTs and competition results in running events between 55% and 85%.

In cycling, a total of eight studies evaluated the relationship between LTs and (simulated) cycling time trial performance (table IV).^[12,89,140,141,165-168] Only one study analysed the correlation with short-duration time trial performance (4000 m individual pursuit) and found a high correlation coefficient of $r=0.86$ in 18 male high-performance track cyclists.^[167] Four studies evaluated distances between 13.5 and 20 km or time trial durations between 20 and 30 minutes.^[89,165,166,168] The correlation coefficients in these studies were in most cases higher (between 0.8 and 0.9) than for the longer time trials (40 km or 60–90 minutes, $r\sim0.7$).^[140,141,165] Overall, the results of these studies were more heterogeneous. Correlation coefficients between LTs and (simulated) competition performance varied between $r=0.23$ ^[165] and $r=0.93$.^[89] In total, the results of the analysed studies point to a common variance of LTs and competition results between 35% and 85% in cycling events. However, the low number of studies and the heterogeneous results point to the need for further carefully designed studies to

arrive at more comprehensive conclusions with regard to the relationship of LTs and time trial performance in cycling.

Two studies were found that analysed the relationship of LT markers with mountain bike cross-country race performance.^[169,170] Such races are usually conducted on a graded terrain with considerable time spent ascending and descending. Impellizzeri et al.^[170] observed high correlations between LT_{Aer} as well as OBLA and race time during a 31 km mountain bike race. Whereas correlations were about 0.7 when LT was expressed in absolute terms, correlations became considerably higher (~0.9) when power output at LT was expressed relative to body mass. Similarly, Gregory et al.^[169] reported higher correlations between LT_{Aer} and a cross-country time trial in 11 male mountain bikers when LT_{Aer} was expressed as related to body mass ($r\sim0.5$ in absolute terms vs $r\sim0.8$ relative to body mass). This finding can be explained with the considerable influence of bodyweight and body composition on performance capacity in cyclists during ascents.^[171-173]

In addition to the studies in running and cycling, another four studies were detected that evaluated LTs and (simulated) competition

Table IV. Correlation coefficients between lactate thresholds and cycling time trial events over various distances and times

Threshold concept	4 km		13.5–20 km; 20–30 min		40 km; 60–90 min	
	PO	VO ₂	PO	VO ₂	PO	VO ₂
LT_{fix}			0.23 ^[165] 0.82 ^[166] 0.90 ^[166]		0.54 ^[165] 0.60 ^[141] 0.81 ^[140]	
Median (min–max)			0.82 (0.23–0.90)		0.60 (0.54–0.81)	
LT_{Aer}	0.86 ^[167]		0.67 ^[165] 0.88 ^[166] 0.86 ^[166] 0.91 ^[168] 0.88 ^[168]		0.91 ^[165] 0.59 ^[141] 0.61 ^[140] 0.69 ^[140] 0.65 ^[140]	0.93 ^[12]
Median (min–max)	0.86		0.88 (0.67–0.91)		0.65 (0.59–0.91)	0.93
LT_{An}			0.45 ^[165] 0.89 ^[166] 0.91 ^[166] 0.93 ^[89]		0.77 ^[165] 0.58 ^[141] 0.52 ^[141] 0.72 ^[141] 0.84 ^[140] 0.83 ^[140]	
Median (min–max)			0.90 (0.45–0.93)		0.75 (0.52–0.84)	

LT_{Aer} = aerobic threshold; LT_{An} = anaerobic threshold; LT_{fix} = fixed lactate threshold; PO = power output; VO₂ = oxygen uptake.

performance. Two of these studies analysed competitive race walkers. Yoshida et al.^[174] found correlation coefficients for OBLA as well as for LT_{Aer} of 0.94 and 0.85, respectively, with walking pace during a 5 km road race in eight female race walkers. Similar results were observed by Hagberg and Coyle^[111] in a heterogeneous group of race walkers with correlation coefficients of 0.94 and 0.82 for velocity and oxygen uptake at LT_{Aer} in a 20 km race walking performance.

Two studies dealt with rowing performance and LTs. Whereas Ingham et al.^[175] observed high correlations ($r=0.86-0.92$) between work rate at fixed and aerobic LTs and 2000 m ergometer performance in 41 rowers of different categories, Cosgrove et al.^[176] found considerably lower correlations ($r=0.39-0.73$) in 13 male rowers.

To summarize, the overwhelming majority of published studies on the relationship between LTs and endurance performance showed strong correlations, particularly for running events. This supports findings of earlier training studies that found training-induced improvements in competitive performance significantly correlated with improvements in LTs.^[130,162] Although it seems likely that other influences such as central nervous system processes may have regulatory and decisive characteristics in endurance events as it was recently claimed,^[177] peripheral metabolic adaptations highly related to the LT^[46] seem to be a necessary and important prerequisite for aerobic endurance performance.

4.3 Lactate Thresholds and Maximal Lactate Steady State

MLSS determination has become very popular in performance diagnosis in several endurance sports. Thus, numerous studies have dealt with the problem of an adequate estimation of MLSS during one single laboratory visit. For instance, some authors tried to estimate MLSS from performance during all-out time trials (5 km or 40 km)^[114,178] from physiological strain (bLa, heart rate, ratings of perceived exertion) during standardized sub-maximal constant-load exercise^[179-182] or from gas exchange measurements.^[183-189]

However, an overview of those studies is beyond the scope of the present review.

There are several studies that examined the metabolic responses during steady-state exercise intensities related to LTs but did not analyse exercise intensities slightly above or below. Schnabel et al.^[190] observed average steady-state lactate concentrations (~ 4.5 mmol/L) during 50-minute runs at the IAT according to Stegmann et al.^[88] However, no other intensity was analysed in this investigation. Stegmann and Kindermann^[146] compared 50-minute cycling exercise in 19 subjects at the IAT as well as at LT4 and found steady-state lactate levels (~ 4 mmol/L) during IAT trials, whereas exercise at LT4 resulted in continuously rising bLa (up to 9.6 mmol/L) and a premature cessation. This is in line with findings of Oyono-Enguelle et al.,^[191] who similarly reported no lactate steady state in three out of five subjects during exercise at LT4. In contrast, Loat and Rhodes^[189] found continuously increasing bLa (on average from 3.4 mmol/L after 15 minutes to 4.6 mmol/L after 45 minutes) and premature fatigue during 60-minute constant load trials at the IAT. However, those authors did not use the originally described test protocol and Heck^[50] has shown that IAT determination is dependent on the protocol used.

Baldari and Guidetti^[144] compared steady-state running at their IAT determined when lactate values were plotted against the corresponding exercise intensity (IAT_m) and against the preceding intensity (IAT_a) and found steady-state lactate levels for IAT_a (~ 4 mmol/L⁻¹) but not for IAT_m . However, due to the determination procedure, the difference between both thresholds was exactly one stage increment and no other intensities in between were evaluated. Ribeiro et al.^[192] assessed a 40-minute steady-state cycling exercise at LT_{Aer} , between LT_{Aer} and LT_{An} (LTP), at LT_{An} as well as between LT_{An} and maximum. Those authors found on average steady-state lactate levels up to LT_{An} (~ 5 mmol/L⁻¹), whereas at the highest intensity, bLa increased continuously and exercise had to be terminated prematurely.

Bacon and Kern^[193] and Tegtbur et al.^[145] compared constant load trials at LMS and 5% or 0.2 m/s, respectively, above the LMS. Those

authors found that LMS intensity but not the higher intensity on average resulted in a lactate steady state. However, in the study of Bacon and Kern,^[193] the average blood lactate increase between minutes 12 and 28 during the constant load trial at the LMS +5% intensity was 1.2 mmol/L, and in four out of ten subjects a lactate steady state according to the recommended criterion^[72,115] was present.

A total of 11 studies evaluated the relationship between one or more LT concepts and MLSS using the recommended procedure, including several constant load trials of at least 30 minutes' duration to determine the MLSS (table V). One study determined MLSS with 20-minute constant load trials.^[113]

Most researchers analysed the relationship of LT4 with MLSS.^[49,72,90,92,112,117] For instance, Heck and colleagues^[49,50,72] found strong correlations between LT4 and MLSS during running as well as during cycling exercise. However, the fitness level of their subjects was quite heterogeneous and, therefore, the high correlations to some extent might be spurious. Additionally, they observed that the velocity at LT4 was higher than MLSS velocity when stage duration during the GXT was 3 minutes, whereas this was not the case with 5-minute stages. Therefore, these authors concluded that LT4 gives a valuable estimate of the MLSS when stage duration is at least 5 minutes. Also, Jones and Doust^[112] found a high correlation between LT4 and the MLSS in a homogenous group of trained runners with LT4 being higher than MLSS (3-minute stages). Lower correlations were found by van Schuylenbergh et al.^[92] in elite cyclists as well as by Beneke^[117] in a homogenous group of rowers. Also, LT4 and MLSS did not differ significantly with 6-minute stages,^[92] whereas LT4 was considerably higher than MLSS with 3-minute stages.^[117] Lajoie et al.^[90] evaluated whether the intensity corresponding to 4 mmol/L lactate during a GXT with 8-minute stages and 30 W increments is appropriate to estimate the MLSS in nine cyclists. Average power output at MLSS and LT4 was not significantly different. However, because bLa at MLSS differed considerably between subjects, the authors concluded

that it is unrealistic to rely on a blood lactate value of 4 mmol/L as a universal criterion for MLSS. Unfortunately, a more detailed analysis regarding the correlation or individual differences between LT4 and MLSS was not reported.

Heck et al.^[49,50] observed high correlations between MLSS and the IAT according to Stegmann et al.^[88] In addition, running velocity was not significantly different between IAT and MLSS independent of stage duration (3 or 5 minutes), whereas in cycling IAT was about 8% higher than MLSS. Urhausen et al.^[86] found in runners as well as in cyclists that constant load trials at IAT resulted on average in a lactate steady state, whereas a 5% higher intensity led to a continuous rise in bLa. Similarly, McLellan and Jacobs^[91] arrived at the conclusion that the IAT is a valid estimate for the MLSS in most subjects, although there exists a considerable difference in a few cases. Unfortunately, these studies reported no measure of correlation between IAT and MLSS or no quantitative data on individual differences between IAT and MLSS. In contrast to the previously mentioned studies, Beneke^[117] found the IAT to be considerably higher than MLSS in nine rowers. Additionally, the correlation in this study was lower than was observed by Heck et al.^[49] This finding might be due to the more homogenous performance level of the rowers as well as to the slow increment in the chosen test protocol.^[50]

Heck et al.^[49] and Heck^[50] found high correlations between the IAT according to Keul et al.^[96] and Bunc et al.^[143] and the MLSS in running and cycling. However, the high correlations might be partly accounted for by the heterogeneous endurance level of the subjects. Furthermore, both thresholds were dependent on the test protocol during the running tests (3-minute vs 5-minute stages).

The LMS was evaluated in two studies.^[89,112] The results of these studies were contradictory. Jones and Doust^[112] found only a low correlation between LMS and MLSS. Additionally, LMS was considerably lower than MLSS. In contrast, LMS was not significantly different from MLSS in the study of MacIntosh et al.^[89] These contrasting observations might have been due to

Table V. Comparison of lactate threshold concepts with MLSS determined by several constant load trials of different intensity

Threshold concept	Subjects	Main outcome	Reference
LT4, OBLA	16 healthy males (running)	High correlation between LT4 and MLSS ($r=0.98$) LT4 on average 0.12 m/s higher than MLSS with 3 min stages but not with 5 min stages during GXT Heterogenous endurance level	Heck et al. ^[49,72]
	22 healthy subjects (cycling)	Significant correlation between LT4 and MLSS ($r=0.92$) LT4 on average 19.9 W higher than MLSS Heterogenous endurance level, slow increase in power output (+6 W/min)	Heck ^[50]
	8 trained male runners	High correlation ($r=0.93$) between OBLA and MLSS OBLA on average 0.4 km/h higher than MLSS	Jones and Doust ^[112]
	21 elite cyclists	Low correlation ($r=0.71$) between LT4 and MLSS No significant difference between LT4 and MLSS (MLSS 15 W higher) Homogenous endurance level	Van Schuylenbergh et al. ^[92]
	9 male rowers	Significant correlation ($r=0.82$) between LT4 and MLSS LT4 significantly higher (32 W) than MLSS Homogenous endurance level	Beneke ^[117]
	10 well trained cyclists	Average power output at LT4 and MLSS was not significantly different (282 W vs 277 W) Strong MLSS criterion (<0.75 mmol/L from 10–60 min) No further data on correlations or intraindividual differences between LT4 and MLSS	Lajoie et al. ^[90]
IAT (Stegmann et al. ^[88])	16 healthy males (running)	High correlation between IAT and MLSS ($r=0.96-0.98$) IAT velocity on average similar to MLSS for 3 min as well as 5 min stages during GXT Heterogenous endurance level of subjects	Heck et al. ^[49]
	22 healthy subjects (cycling)	Significant correlation between IAT and MLSS ($r=0.87$) IAT on average 15.1 W higher than MLSS Heterogenous endurance level, slow increase in power output (+6 W/min) not corresponding to the originally described test protocol	Heck ^[50]
	16 trained cyclists	CLT at and below IAT resulted on average in LSS but not CLT at 105% IAT	Urhausen et al. ^[86]
	14 trained runners	100% IAT does not in all individuals exactly represent MLSS LSS was found in 6 (of 14 runners) and 9 (of 16 cyclists) at 105% IAT No further data on correlations or intraindividual differences between IAT and MLSS	
	11 males (cycling)	CLT at LT4 (cycling at 104% IAT) resulted on average not in a LSS No LSS during CLT at IAT +5% $\dot{V}O_{2max}$; only 1 LSS during CLT at IAT +2.5% $\dot{V}O_{2max}$ Two subjects showed no LSS during CLT at IAT -7.5% $\dot{V}O_{2max}$, all other subjects showed LSS during CLT at IAT -2.5% $\dot{V}O_{2max}$ No further data on correlations or intraindividual differences between IAT and MLSS	McLellan and Jacobs ^[91]
	9 male rowers	Significant correlation ($r=0.81$) between IAT and MLSS IAT significantly higher (32 W) than MLSS	Beneke ^[117]
IAT (Keul et al. ^[96])	16 healthy males (running)	High correlation between IAT and MLSS ($r=0.98$) IAT velocity on average 0.2 m/s higher than MLSS with 3 min stages and slightly lower with 5 min stages during GXT Heterogenous endurance level of subjects	Heck et al. ^[49]
	22 healthy subjects (cycling)	Significant correlation between IAT and MLSS ($r=0.94$) IAT on average 21.0 W higher than MLSS Heterogenous endurance level, slow increase in power output (+6 W/min ⁻¹)	Heck ^[50]

Continued next page

Table V. Contd

Threshold concept	Subjects	Main outcome	Reference
IAT (Bunc et al. ^[143])	16 healthy males (running)	High correlation between IAT and MLSS ($r=0.98-0.99$) IAT velocity on average considerably higher than MLSS for 3-min (+0.31 m/s) as well as 5 min stages (+0.14 m/s) during GXT Heterogenous endurance level of subjects	Heck et al. ^[49]
	22 healthy subjects (cycling)	Significant correlation between IAT and MLSS ($r=0.89$) IAT on average 71.5 W higher than MLSS Heterogenous endurance level, slow increase in power output (+6 W/min)	Heck ^[50]
LMS	10 trained male runners	Low correlation ($r=0.61$) between LMS and MLSS LMS on average 0.8 km/h lower than MLSS	Jones and Doust ^[112]
	14 cyclists or triathletes	LMS on average not different from MLSS No good estimate of MLSS by LMS in three subjects MLSS criterion: <0.7 mmol/L during last 20 min No further data on correlations or intraindividual differences between LMS and MLSS	MacIntosh et al. ^[89]
D_{mod}	21 elite cyclists	Significant correlation ($r=0.85$) between D_{mod} threshold and MLSS D_{mod} threshold significantly lower (-23 W) than MLSS	Van Schuylenbergh et al. ^[92]
LTP	8 males (running)	No correlation between LTP and MLSS ($r=0.18$) On average no difference between LTP and MLSS (13.7 vs 13.8 km/h) 95% LoA ^[194] = ± 1.8 km/h	Smith and Jones ^[103]
LT_{Aer}	10 trained male runners	High correlation ($r=0.94$) between LT_{Aer} and MLSS LT_{Aer} on average 0.6 km/h lower than MLSS	Jones and Doust ^[112]
	11 male recreational runners	No correlation of LT_{Aer} with MLSS (speed: $r=-0.01$; $\dot{V}O_2$: $r=-0.47$) LT_{Aer} on average 1.1 km/h lower than MLSS 20 min CLT, but strong MLSS criterion (<0.2 mmol/L)	Haverty et al. ^[113]

CLT=constant load trial; D_{mod} =maximal distance from blood lactate concentration (bLa) curve to the line formed by the point before the first rise in bLa and the value at cessation of exercise; GXT=incremental exercise test; IAT=individual anaerobic threshold; LMS=lactate minimum speed; LoA=limits of agreement; LSS=lactate steady state; LT4=4 mmol/L lactate threshold; LT_{Aer} =aerobic threshold; LTP=lactate turnpoint; MLSS=maximal lactate steady state; OBLA=onset of blood lactate accumulation; r =correlation coefficient; $\dot{V}O_{2max}$ =maximal oxygen uptake.

the considerably different test protocols used in both studies. This is in line with the findings of Carter et al.^[148] showing that LMS is highly dependent on the test protocol.

For other threshold concepts, scientific data regarding the relationship of the threshold and MLSS are scarce. Van Schuylenbergh et al.^[92] found a significant correlation between the D_{mod} -threshold and MLSS, although D_{mod} was significantly lower than MLSS. In contrast, LTP was found to be not different from MLSS on average, but it was not correlated to MLSS and the 95% limits of agreement (LoA)^[194] of the difference between LTP and MLSS were wide.^[103]

There were also two studies that analysed the relationship between MLSS and LT_{Aer} .^[112,113] As could be expected, LT_{Aer} was situated considerably below the MLSS in both studies. Whereas Jones and Doust^[112] reported a high

correlation between LT_{Aer} and MLSS, Haverty et al.^[113] did not. This might be due to short constant load trials (20 minutes) and the strict MLSS criterion (<0.2 mmol/L increase during the last 10 minutes) in the latter study, which does not sufficiently consider the time course of bLa changes and may have led to an underestimation of the real MLSS.^[115]

To summarize, there is evidence that some LT concepts might be able to estimate the MLSS. In particular, the IAT according to Stegmann et al.,^[88] and LT4 were repeatedly examined. Mostly linear regressions or average lactate courses were reported. Correlations and regressions determine relative reliability of two methods but do not assess systematic bias or absolute agreement. Furthermore, they depend greatly on the range of values in the analysed sample.^[195] Thus, from a practical and statistical point of

Table VI. Mean bias (difference maximal lactate steady state [MLSS]-LT) and 95% limits of agreement (LoA) for four different lactate threshold concepts during treadmill (n = 16) and cycle ergometry (n = 22). Results calculated from raw data reported by Heck et al.^[49,50,72] (with permission)

Lactate threshold concept	Treadmill ergometry 3 min stages, +0.4 m/s			Treadmill ergometry 5 min stages, +0.4 m/s			Cycle ergometry 2 min stages, +25 W		
	mean bias (m/s)	LoA (m/s)	LoA (%)	mean bias (m/s)	LoA (m/s)	LoA (%)	mean bias (W)	LoA (W)	LoA (%)
LT4	-0.13	±0.35	±8	0.02	±0.39	±9	-19.8	±28.4	±14
IAT (Keul et al. ^[96])	-0.20	±0.39	±9	0.06	±0.35	±8	-21.0	±22.4	±11
IAT (Stegmann et al. ^[88])	-0.03	±0.51	±12	-0.03	±0.37	±9	-15.0	±35.0	±18
IAT (Bunc et al. ^[143])	-0.33	±0.33	±8	-0.14	±0.37	±9	-71.4	±52.8	±27

IAT = individual anaerobic threshold; LT4 = 4 mmol/L threshold.

view it would be of interest to know the absolute variability of individual differences between the LT and MLSS. An appropriate means to report this variability may be the mean bias and the 95% LoA as it was described by Bland and Altman.^[194] There is only one study available that applied this procedure.^[103] Such a procedure would also allow for assessing heteroscedasticity (i.e. whether the differences depend on the magnitude of the mean or – in this case – endurance capacity).^[195]

Table VI shows an example calculation of the mean bias and the 95% LoA for four different LT concepts from raw data reported by Heck et al.^[49,50,72] These data show a mean bias between 0.5% and 8%, with LoA of about 10% in a running exercise. This means that for each new subject within the study population it could be expected (with a 95% probability) that the difference between MLSS and the respective LT is within these LoA.^[195] For the cycling exercise the results are more heterogenous with greater mean bias and LoA. However, due to the limited data points these observations are preliminary and should be confirmed by further research.

5. Conclusions and Perspectives

In conclusion, it can be stated that a huge amount of evidence exists that LT concepts are of considerable importance for the diagnosis as well as the prediction of aerobic endurance performance. The concept of the aerobic-anaerobic transition may serve as a reasonable means for

performance diagnosis and intensity prescription in endurance sports. However, there are several open questions that should be appropriately addressed by future research. These are:

- Whereas the relationship of LTs with competition performance is well established in running events and less strongly in cycling, there is lack of evidence for most other endurance sports.
- Scientific studies comparing LTs with MLSS are rare and the results are partially conflicting. This might be due to different methodological approaches. It is suggested that the MLSS be assessed by the established procedure using several constant load trials with different intensities^[72,115] and that the MLSS be compared with a chosen LT. To do so, measures of absolute agreement between LTs and MLSS should be reported according to the method introduced by Bland and Altman.^[194]
- In this context, it is important to know the basic variability and reproducibility of the MLSS. Up to now, no scientific data addressing this question exist. Therefore, it is recommended to evaluate the variability of MLSS in future research. Of note, this may enable an evaluation of the differences between LT and MLSS compared with the basic variability of the MLSS and, thus, give more detailed information on the quality of the MLSS estimate.

Although there has been much and controversial debate on the LT phenomenon during the last three decades, many scientific studies have dealt with LT concepts, their value in as-

sessing endurance performance or in prescribing exercise intensities in endurance training. It might be speculated that a considerable part of the debate has to be attributed to the misinterpretation of the physiological basis of the phenomenon. The presented framework may help to clarify the controversy and may give a rational basis for performance diagnosis and training prescriptions in future research as well as in sports practice.

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